Explainable AI Clinical Decision Support System

Clinical Prediction Report

Condition: Stroke

Generated On: 14-09-2026 02:14:32

Patient Information

Field	Value
Patient ID	PAT003
Patient Name	Final Test Patient
Doctor / Clinician	Dr. Smith
Hospital / Clinic	Fortis Hospital
Disease	Stroke

Clinical Parameters

Feature	Observed Value
Gender	Male
Age	12
Hypertension	0
Heart Disease	0
Ever Married	No
Work Type	children
Residence Type	Urban
Average Glucose Level	120
Body Mass Index	12
Smoking Status	never smoked

Model Assessment

Field	Value
Condition	Stroke
Prediction	No Stroke
Class	No Stroke
Probability	4.43%

Field	Value
Risk Level	Low
Clinical Recommendation	The model indicates a lower likelihood of stroke risk. This result should still be interpreted together with clinical assessment and professional judgement.

SHAP Evidence

Feature contribution to the model decision.

Feature	Contribution
Age	-2.7830
Work Type Children	0.7235
Smoking Status Never Smoked	-0.1224
Body Mass Index	-0.1090
Hypertension	-0.0956

LIME Evidence

Local explanation for this patient.

Feature / Rule	Weight
age <= -0.77	-0.4249
work_type_Never_worked <= -0.06	0.3291
work_type_children > -0.40	0.1129
avg_glucose_level > 0.17	0.0654
hypertension <= -0.33	-0.0635

SHAP-LIME Agreement

Consistency between the SHAP and LIME local explanation methods.

Field	Value
Agreement Score	76.00%
Feature Overlap	40.00%
Direction Agreement	100.00%
Ranking Agreement	100.00%
Overall Consistency	76.00%
Consistency Level	Moderate
Shared Feature Count	2

Field	Value
Shared / Common Features	Age, Htn
SHAP Only	Work Type Children, Smoking Status Never Smoked, Bmi
LIME Only	Work Type Never Worked, Work Type Children, Avg Glucose Level

Agreement measures consistency between SHAP and LIME explanations. It indicates explanation consistency, not clinical certainty.

Multi-Feature Counterfactual Explanation

Counterfactual explanations identify hypothetical alternative patient profiles in which one or more clinical features are changed to determine whether the model prediction can change.

Field	Value
Original Prediction	No Stroke
Original Probability	4.43%

No counterfactual profile was found.

The model did not return a valid alternative profile for this stroke assessment within the configured search space.

Counterfactual explanations are hypothetical model-based scenarios. They describe changes that may alter the model output and must not be interpreted as treatment recommendations, prescriptions, or required patient targets.

Clinical Interpretation Note

The prediction and explanation values in this report represent the output of the machine-learning system for the submitted patient data. SHAP and LIME provide model interpretability, while counterfactual analysis illustrates hypothetical changes that may cross the model's decision boundary.

Disclaimer: This report is intended for clinical decision support only. Model outputs, SHAP values, LIME explanations, agreement scores, and counterfactual scenarios must not be interpreted as definitive diagnoses, prescriptions, treatment instructions, or required clinical targets. Final clinical decisions must be made by qualified healthcare professionals using appropriate clinical assessment and additional evidence.

Generated by Explainable AI Clinical Decision Support System

Machine Learning • SHAP • LIME • SHAP-LIME Agreement • Counterfactual Analysis • Flask • React

Doctor Signature

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Dr. Smith

Patient Signature

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Final Test Patient